

□ Assign 8 current components: $I_1 \rightarrow I_8$

$$+V_1 - V_2 - V_3 + V_4 = 0 \Rightarrow 1I_2 - 2I_5 - 3I_6 + 4I_4 = 0 \quad \text{--- (11)}$$

$$+V_7 + V_3 - V_6 = 0 \Rightarrow +7I_7 + 3I_6 - 6I_8 = 0 \quad \text{--- (IV)}$$

$$\text{KCL 2: } -I_5 + I_6 + I_8 = 1 \quad \text{--- (v1)}$$

KCL 3: $-I_2 + I_3 + I_4 = 0$ — VII

KCL 4: $-I_4 - I_6 + I_7 = 0$ — (VIII)

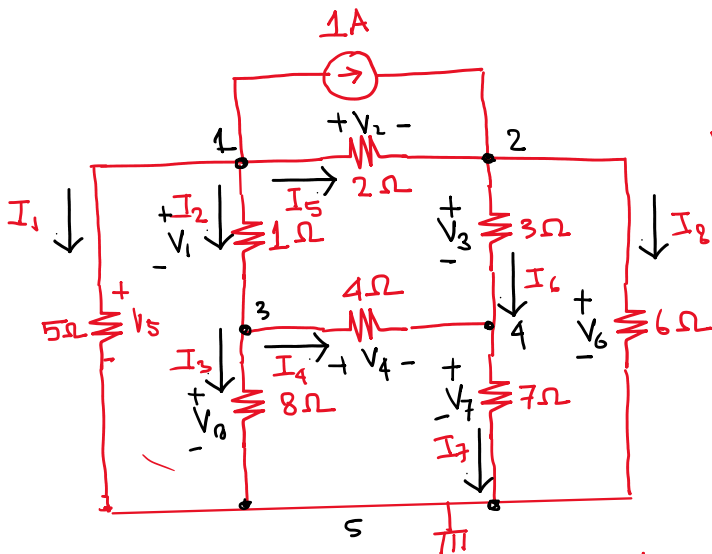
Write as many KCL equations so that each variable is covered at least once.

$$\begin{bmatrix} \vdots \\ I_1 \\ \vdots \\ I_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

Voltage betⁿ 2 nodes ($V_2 - 0$) = Summation of all the voltage drops and rise along each path betⁿ the nodes

Node Analysis: 5 nodes: 4 (Non-Ref) + 1 Ref
4 equations

$$I_S = \frac{V_1 - V_2}{2}$$



$$\Rightarrow V_2 = V_6 = 6 I_8$$

$$\Rightarrow V_7 + V_3 = V_2$$

$$\therefore 7I_7 + 3I_6 = V_L$$

$$5I_1 - 1I_2 - 8I_3 = 0 \quad \text{--- (1)}$$

$$5I_1 - 1I_2 - 8I_3 + 0I_4 + 0I_5 + 0I_6 + 0I_7 + 0I_8 = 0$$

$$\begin{bmatrix} 5 & -1 & -8 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \\ I_6 \\ I_7 \\ I_8 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$